22 x is an integer such that $5x > 38$

(a) Find the least value of x.

$5x > 38$

$x > 38 \div 5 = 7.6$

Least integer = 8

2 marks total
Divide by positive 5.
Next whole number above 7.6.

8

(2)

(b) Solve $3p - 7 = 5p + 10$

$3p - 7 = 5p + 10$

$-17 = 2p$

$p = -8.5$

2 marks total
Subtract 3p and 10;
divide both sides by 2.

$p =$ −8.5

(2)

Tammy has been asked to solve the equation $16h^2 = 400$

Here is her working.

$\sqrt{400} = 20$

$20 \div 16 = 1.25$

Tammy's answer is wrong.

(c) Describe a mistake Tammy has made.

She should divide by 16 before taking the square root.

$16h^2 = 400 \rightarrow h^2 = 25 \rightarrow h = \pm 5$.

1 mark
Taking a square root first also requires square-rooting 16.

(1)

(Total for Question 22 is 5 marks)

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